



compasses were made for navigational purposes. Long back, the compass was known as the “matsya yantra” in our country, because the device consisted of a metallic fish (from an early Chinese design) placed in a cup of oil; the oil was used to make the magnetic needle settle quickly. However, the early compasses didn't really help Columbus in his voyage when he landed at West Indies instead of India. It was found that the true geodetic north (the North Pole) is different from the magnetic north (in northern Canada), as pointed out by the magnetic needle. As time passed, better, more accurate and purpose-specific compasses were made. Gyrocompass is a non-magnetic compass that finds the true north. It's similar to the gyroscope, in that it uses a (electrically powered) fast-spinning disc and friction forces due to rotation of the Earth. This type of compass is mainly used on ships as it's not affected by ferromagnetic metals. Today we use our smartphones to give us directions, possible due to in-built magnetometers. It provides data to the microprocessor and directions are calculated using trigonometry by internal electronics.